



CURRENCY X CHANGE

By Team Anything:

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- Loh YuanChao
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- Ng Wan Ying
- Arora Manav

CurrencyXChange

What is it?

- Web Application
- Find latest currency exchange rates
- Easily accessible



www.meowmeow.icu

Why choose us?

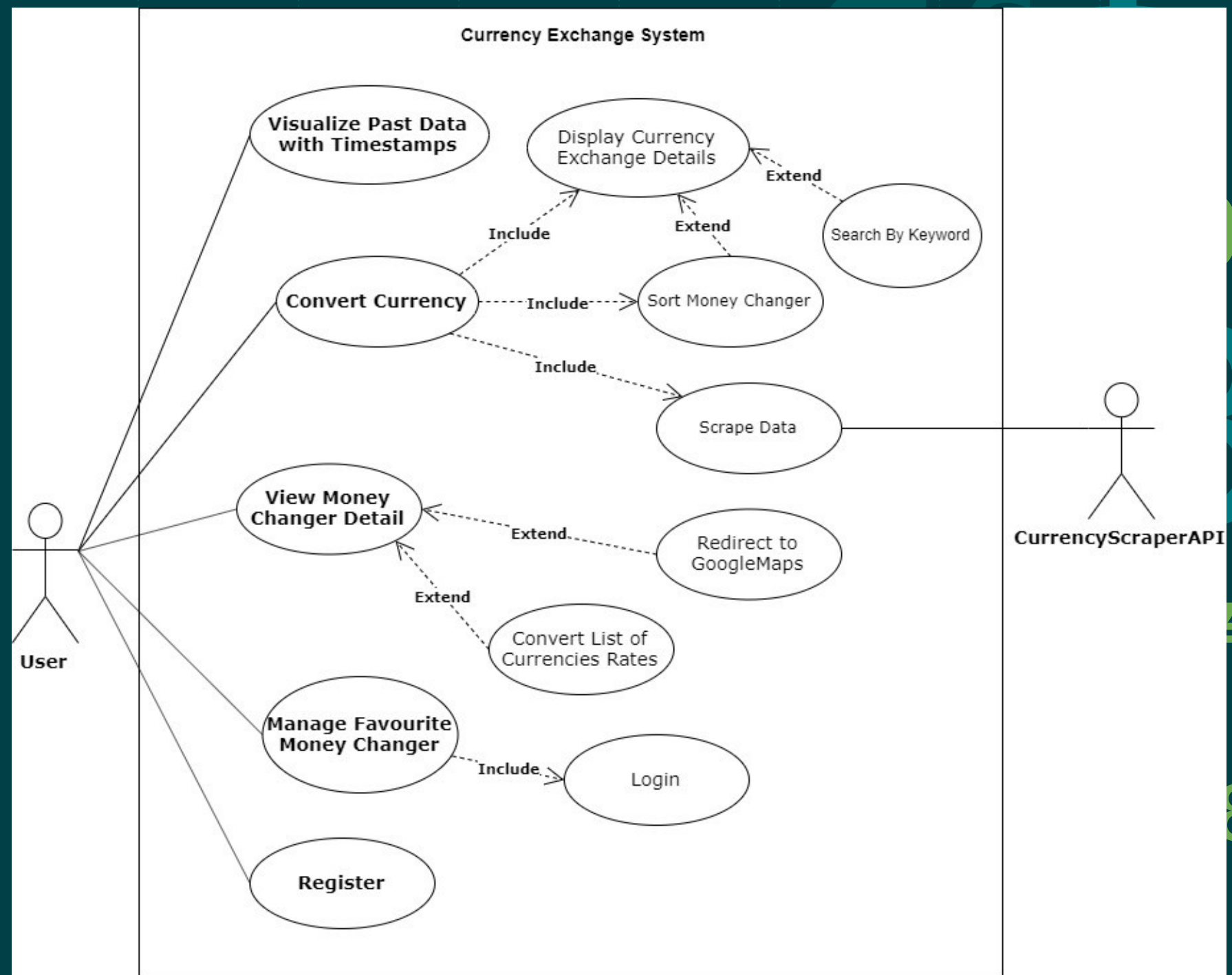
- Find all currency exchange rates in one place

Our Target Audiences:

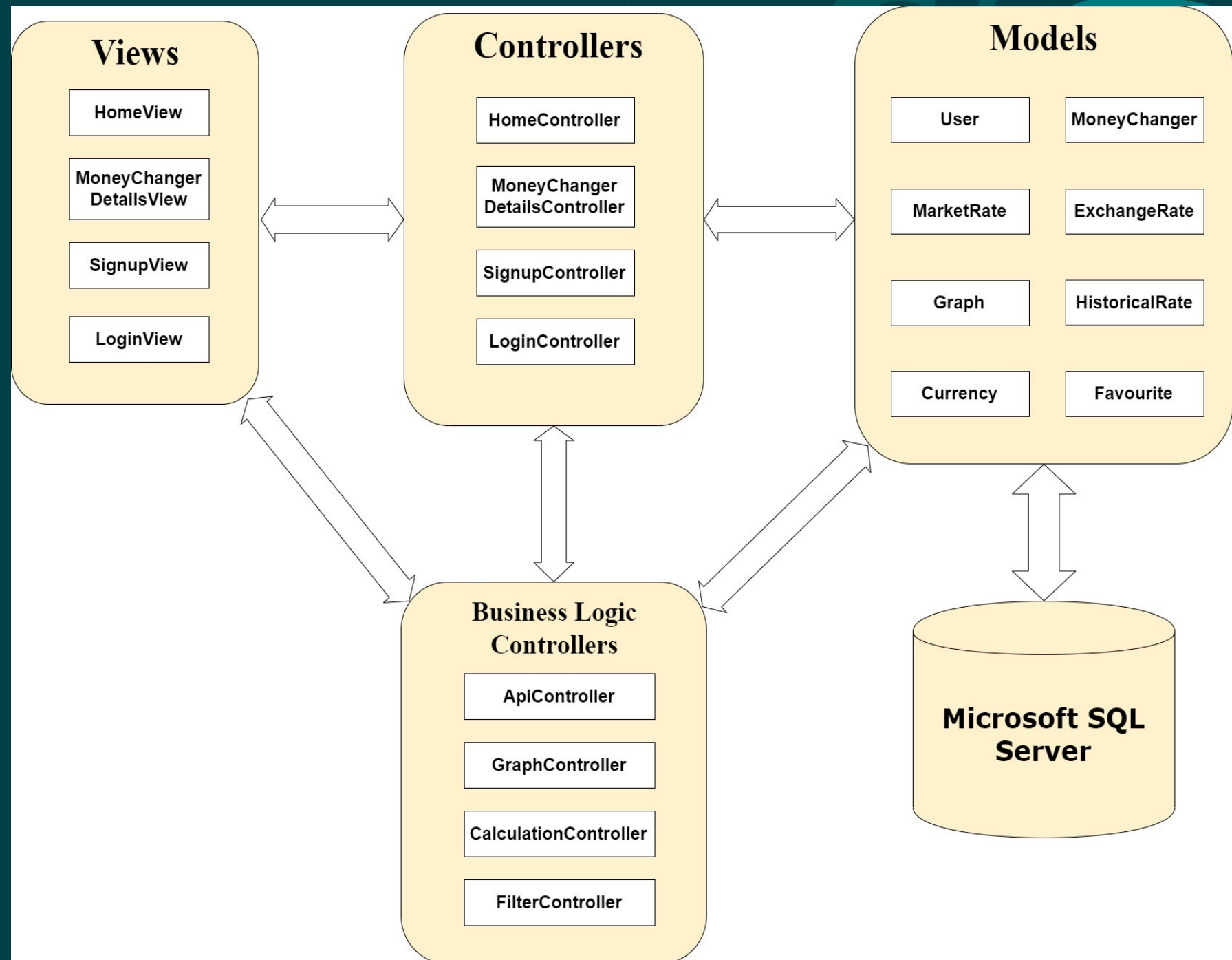
- Travellers
- Tourists
- Importers/Exporters
- Anyone with interest in currency exchange



FEATURES



SYSTEM DESIGN - MODEL VIEW CONTROLLERS



A wide, straight asphalt road stretches into the distance, flanked by snow-dusted hills under a cloudy sky. The road is dark and textured, with a dashed white line down the center. The hills on either side are covered in patches of snow and sparse vegetation. The sky is overcast with soft, grey clouds. The overall mood is quiet and expansive.

DEMO

GOOD TRACEABILITY

-  1. Archive
-  2. SRS
-  3. System Design
-  4. Testing
-  5. Meeting Minutes
-  CZ2006 VIDEO

GOOD TRACEABILITY

Revision History

Name(s)	Date	Reason For Changes	Version
Team	03 Sep 2019	Initial Requirement Elicitation	1.0
Team	08 Sep 2019	Additional System Requirements	1.1
Team	08 Sep 2019	Refine System Requirements To Include Atomicity	1.2
Arora Manav	09 Sep 2019	Initial Purpose, Product Scope, Mission Statement and Product Perspective	1.3
Loh Yuanchao	20 Sep 2019	Modify System Requirements	2.0
Ng Wan Ying	24 Sep 2019	Modify System Requirements	2.1
Loh Yuanchao	30 Sep 2019	Finalizing System Requirements	3.0
Arora Manav	30 Sep 2019	Modifying User Classes and Characteristics	3.1
Arora Manav	12 Oct 2019	Initialising Purpose, Intended Audience and Reading Suggestions, Product Scope, Mission Statement, Product Perspective, Product Functions,	4.1

GOOD TRACEABILITY

Arora Manav	12 Oct 2019	Initial Operating Environment, Design and Implementation Constraints, Assumptions and Dependencies	4.2
Loh Yuanchao	12 Oct 2019	Updates on External Interface Requirements, User Interfaces, Hardware Interfaces, Software Interfaces, Communication Interfaces and other Non-Functional Requirements	4.3
Arora Manav	12 Oct 2019	Modify System Requirements and Initialising UC07-UC10	4.4
Arora Manav	21 Oct 2019	Initialising UC11 and UC12 Modifying Design and Implementation Constraints, Product Scope, User Classes and Characteristics	4.5
Yap Guan Sheng	21 Oct 2019	Executive Summary Provided Project Highlights	4.5
Loh Yuanchao	21 Oct 2019	Initialize UC13	4.5
		Modified UC2	
Loh Yuanchao	27 Oct 2019	Update Data Dictionary	4.6
Arora Manav	27 Oct 2019	Finalising System Requirements	4.6

CONVERT CURRENCY

Actor:	Customer (Initiating)
Description:	The "Convert Currency" Use Case computes the user input amount in the currency of their preference to their desired currency and displays the cash amount that the user will receive after the conversion.
Preconditions:	1. The user entered the system URL into his browser URL bar.
Postconditions:	1. The user gets a list of converted amount offered by various money changers.
Priority:	High
Frequency of Use:	High
Flow of Events:	1. The system use the included UC01 Scrape Data to get the currency data and money changer data 2. The user selects the selling currency. 3. The user enters the amount to sell. 4. The user selects the buying currency and press convert. 5. The system validates the amount.
	6. The system computes currency conversion using current market rate. 7. The system computes currency conversion using the exchange rate. 8. The system uses the included UC04 Sort Money Changer to sort the computed currency conversion result by best exchange rate.
Alternative Flows:	AF-S5: If the system fails to verify the amount to be valid. 1. The system displays a meaningful invalid message. 2. The user acknowledge the unsuccessful outcome. 3. The system returns to Step 2.
Exceptions:	NIL
Includes:	UC01, UC04

CONVERT CURRENCY

Functional Requirements:	2.1 The system must display the market rate for default exchange rates. 2.1.1 The system must be able to access the database with latest market rate. 2.2 The system must allow the user to convert currency. 2.2.1 The system must display a list of currency to select. 2.2.1.1 The display of currency must be sorted according to country name. 2.2.1.2 The display of currency must consist of an image of the country's flag. 2.2.1.3 The display of currency must consist of the currency's market rate. 2.2.2 The system must allow the user to enter an amount to convert. 2.2.2.1 The system must verify that the input amount from user is valid.
	2.2.2.1.1 The input amount must consist of digits only. 2.2.2.1.2 The input amount must not be a negative value. 2.2.2.2 The system must display an invalid message and prompt the user to re-enter a valid amount if the input amount from user is invalid. 2.2.3 The system must compute the amount the user get after conversion. 2.2.3.1 The system must round off the amount to the nearest cent.

CONVERT CURRENCY

Non-Functional Requirements:	1. The system must display the form for user to convert currency within 0.5 seconds. 2. The system must validate the user input amount and perform necessary action after validation within 0.3 seconds. 3. The system must compute the required result on currency conversion that is sorted by best rate within 6 seconds.
Special Requirements:	NIL
Assumptions:	NIL
Notes and Issues:	NIL

CORE IMPLEMENTATIONS

```

1 reference | Zheng Tao, 30 days ago | 1 author, 1 change | 0 exceptions
public static async System.Threading.Tasks.Task SetRates()
{
    //Set api key
    string apiUrl = "http://data.fixer.io/api/latest?access_key=" + Key + "&base=eur";
    //Set rates
    using (var client = new HttpClient())
    {
        var uri = new Uri(apiUrl);
        //wait for request
        var response = await client.GetAsync(uri);
        //Get json data
        string textResult = await response.Content.ReadAsStringAsync();
        JavaScriptSerializer j = new JavaScriptSerializer();
        //Convert json to MarketRate object
        MarketRate a = (MarketRate)j.Deserialize(textResult, typeof(MarketRate));
        rates = a.rates;
    }
}

1 reference | Zheng Tao, 30 days ago | 1 author, 1 change | 0 exceptions
public static async System.Threading.Tasks.Task<MarketRate> GetHistoryRateAsync(DateTime thisDate)
{
    //Set the api url
    string apiUrl = "http://data.fixer.io/api/" + thisDate.ToString("yyyy-MM-dd") + "?access_key=" + ApiController.Key + "&base=EUR";
    //Get all the past 30 days rates
    using (var client = new HttpClient())
    {
        var uri = new Uri(apiUrl);
        //wait for request
        var response = client.GetAsync(uri).Result;
        //Get json data
        string textResult = await response.Content.ReadAsStringAsync();
        JavaScriptSerializer j = new JavaScriptSerializer();
        //Convert json to MarketRate object
        MarketRate marketRate = (MarketRate)j.Deserialize(textResult, typeof(MarketRate));
        return marketRate;
    }
}

1 reference | Zheng Tao, 30 days ago | 1 author, 1 change | 0 exceptions
public static object GetCurrency(float ExchangeAmount, string ExchangeFrom, string ExchangeTo)
{
    //Exchange currency of ExchangeFrom to ExchangeTo
    //Get ExchangeFrom rates
    float exchangeFrom = rates.Where(z => z.Key == ExchangeFrom).FirstOrDefault().Value;
    //Get ExchangeTo rates
    float exchangeTo = rates.Where(z => z.Key == ExchangeTo).FirstOrDefault().Value;
    //Convert the amount
    double amount = CalculationController.ConvertCurrency(ExchangeAmount, exchangeFrom, exchangeTo);
    //Get the base rate as api is in EUR
    double baseRate = CalculationController.ConvertCurrency(1, exchangeFrom, exchangeTo);
    var result = new { Amount = amount, Rate = baseRate };
    return result;
}

```

BLACKBOX TESTING

1. Sign Up Page

1.1 Equivalence Class Testing

Test Parameters	Equivalence Classes (ECs)			
	EC Types	Valid EC	Invalid EC	Invalid EC
Username	Discrete Values	>= 8 characters	< 8 characters	NIL
	Discrete Values	Alphabets and digits	Symbols	NIL
		>= 8 characters of	< 8 characters of	NIL
Password	Discrete Values	alphabets, numbers or symbols	alphabets, numbers or symbols	
Confirm Password	Discrete Values	Same as Password	Different from Password	NIL
Mobile Number	Discrete Values	= 8 digits	!= 8 digits	NIL
	Discrete Values	First digit is 8 or 9	First digit is not 8 or 9	NIL

WHITEBOX TESTING

1

```
thisMC = model.MoneyChangers.Where(z => z.Name == Name).FirstOrDefault()
```

2

if thisMC == null

True

3

```
thisMC = new MoneyChanger()
```

4 - 5

```
var arr = Address.Split(" ")  
thisMC.Name = Name
```

6

if arr.Count() > 0

True

7 - 9

```
string PostalCode = arr[arr.Count() - 1]  
thisMc.Location = Address.Replace(" ", " + PostalCode, "")  
thisMc.PostalCode = PostalCode
```

10 - 12

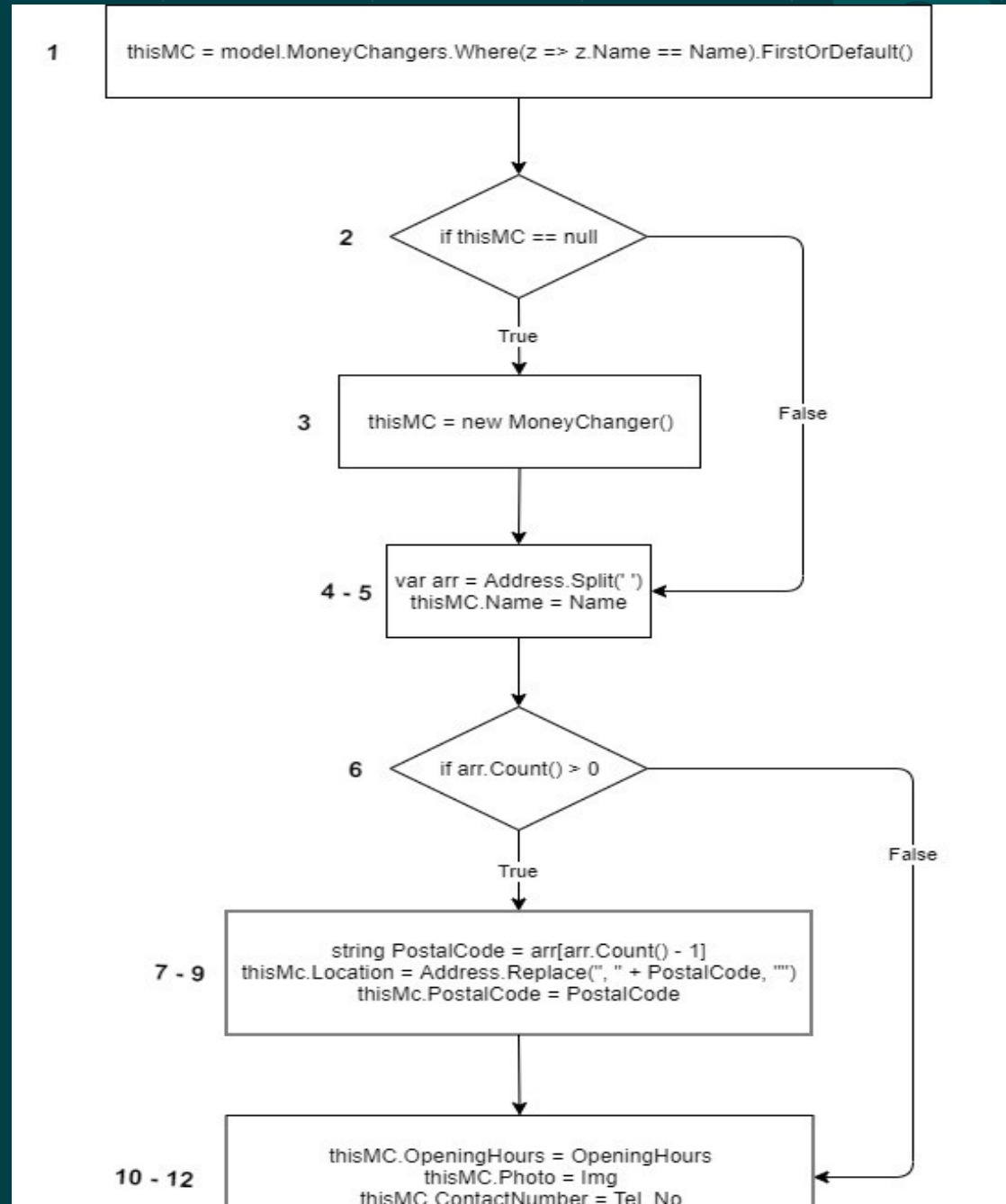
```
thisMC.OpeningHours = OpeningHours  
thisMC.Photo = Img  
thisMC.ContactNumber = Tel_No
```

False

False



WHITEBOX TESTING



Good SE Practices



**For online
collaboration**

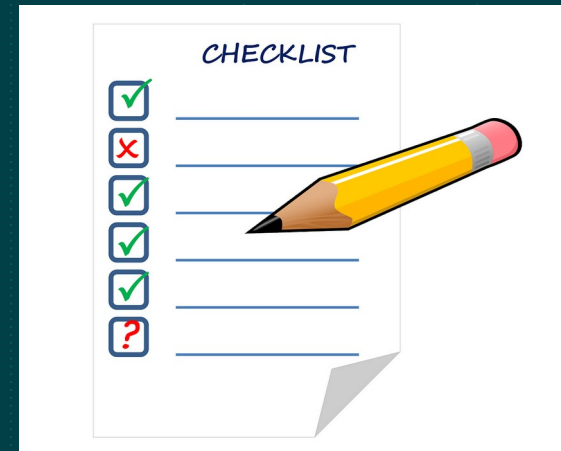


**Software Configuration
Management**

Good SE Practices

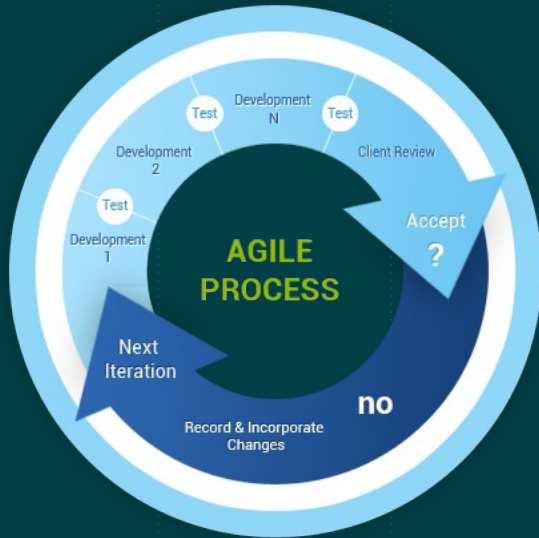


**Consistent
Documentation**



Traceability

Good SE Practices



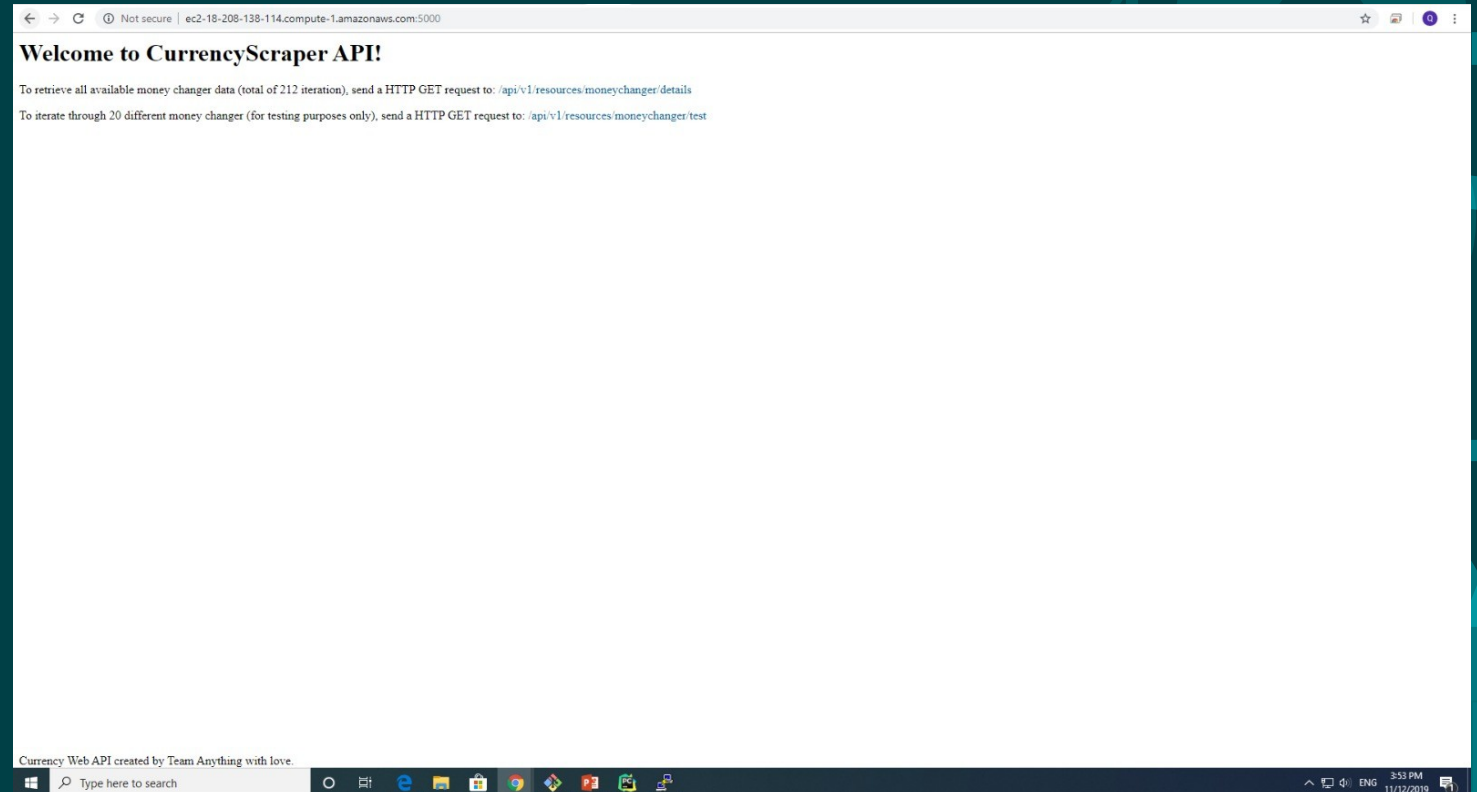
**Agile
Methodology**



**Good Project
Practices**

Good SE Practices

- CurrencyScraperAPI built with the following in mind:
 - Flexibility
 - Scalability
 - Extensibility
 - Reusability



Good SE Practices

- CurrencyXChange app built with MVC architecture that promotes:
 - Loose coupling
 - High cohesion



Challenges Faced

- Initial Diverse Opinions
- Time Coordination
- Discovering & Fixing of Bugs



Thank You!

The End



Credits

- <https://thenounproject.com/term/computer/116/>
- https://www.flaticon.com/free-icon/android_174836
- <https://www.uihere.com/free-cliparts/apple-ii-logo-color-rainbow-sticker-1563822>
- <https://www.anchor.com.au/>
- <https://www.google.com/drive/>
- <https://www.usherwood.com/managed-documents>



Credits

- <https://www.pngguru.com/free-transparent-background-png-clipart-ecxca>
- https://docs.google.com/presentation/d/1v5ISr1mFKUkiCDtwSbl2g9Ae0mts4DynuB-hluq5sms/edit#slide=id.g35f391192_057

